

Supplementary materials for: A Multi-Stream Temporal Context Model for narrative memory

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S0. Architectural lineage of MS-TCM

The present paper describes MS-TCM as a multi-stream extension of Cornell and Zhang^{S1} hierarchical CMR. The model evolved through several preceding architectural revisions whose details are recorded in the iteration log at `notes/mstcm.architecture.log.md` and are not part of the present specification. The earliest formulation (`notes/ms-tcm.pdf`) used a single global context plus a bank of storyline-specific context vectors mixed into an encoding composite with fixed weights, plus a task-dependent retrieval reweighting; that architecture failed to reproduce the canonical free-recall phenomena (serial position curve, probability of first recall, lag-CRP) because it bypassed the standard CMR retrieval route, and was retired in favour of the v6 design (`notes/two_level_cmr_v6.pdf`) which rebases MS-TCM on top of Cornell and Zhang^{S1}. Subsequent empirical observations in the FRFR-category worked example motivated two further additions to the v6 design: a τ recall-onset mixture (because the C&Z 2025 baseline structurally cannot produce a primacy spike in the probability of first recall) and a strict hierarchical recall route with a separate global cross-storyline context (because participants organize recall by category even when the presentation order is randomized). The present paper consolidates those additions into a single MS-TCM specification — the model described in §2 of the main text and implemented at `code/ms_tcm/_likelihood_core_mstcm.py`. The iteration-by-iteration log of empirical motivations and architectural decisions is at `notes/mstcm.architecture.log.md`; the symbol inventory and file-level deletion manifest from the earlier-spec migration is at `notes/v6.migration.md`; git history preserves earlier sources.

S1. Off-by-one correction in the reformatted recall table

The upstream `exp2.egg` file from^{S3} encodes each recall event's provenance in a `temporal` HDF5 attribute that is 0-indexed against the `i{k}` subgroup names in the presented group. The initial reformat (`scripts/reformat_frfr_category.py`, commit `8684fa0`) treated `temporal` as if it were 1-indexed, which shifted every in-list recall's serial position down by one and silently re-coded the last word of every list as an extra-list intrusion. The corrected mapping (commit `405e272`, `sp =`

`temporal + 1` for `temporal` $\in [0, 15]$, else `sp = 0`) restores the expected serial-position distribution (primacy at SP1–2, recency at SP14–16) and reduces the extra-list intrusion count from 399 to 10.

A regression test at `code/tests/test_frfr.py::test_recall_serial_position_shape` asserts $P(\text{recall}) > 0.3$ for the last three serial positions and an intrusion rate below 5%, so any future off-by-one would fail CI.

S2. Likelihood-surface flatness on small synthetic data

At FRFR-scale real data the MS-TCM likelihood surface is informative (grid scans across the active parameters $(\beta_{\text{enc}}, \gamma_{\text{ic}})$ give a range of ~ 100 nats across the parameter simplex, vs a uniform-recall baseline ~ 1200 nats below the MLE). At the smaller scales used for quick sanity checks, the same grid scan shows an essentially flat surface (LL varies by ~ 2 nats around a uniform baseline). The practical consequence is that parameter-recovery tests need a synthetic dataset at FRFR scale ($30 \text{ pt} \times 16 \text{ lists} \times 16 \text{ words} \times 4 \text{ categories} \approx 7,680$ recalls) before the L-BFGS-B optimizer can reliably recover the true parameter vector; smaller datasets converge to whatever corner the initial point leans toward. This is a property of the likelihood surface, not a bug in the fitter.

S3. Implementation details: norm-preserving drift, softmax gain, primacy, no-repeats

Four implementation choices beyond the minimal equations in the main text are necessary for the model’s forward simulator and its posterior-predictive to match the observed behavioural regularities (SPC, P(first recall), lag-CRP, clustering). Each is standard in the CMR / hierarchical-CMR literature but is worth stating explicitly.

Unit-norm feature encoder. Equation (1) of the main text assumes unit-norm inputs so the drift invariant $\|\mathbf{c}(t)\| = 1$ is preserved at every step. Raw multi-hot feature vectors (with k active slots) have norm $\sqrt{k}$; the feature encoder therefore L_2 -normalises each row by default. See `code/ms_tcm/features.py (encode_features(..., normalize=True))`.

Norm-preserving ρ under non-orthogonal inputs. For strictly orthonormal inputs, Howard and Kahana^{S2}’s $\rho = \sqrt{1 - \beta^2}$ preserves $\|\mathbf{c}(t)\| = 1$ exactly. Multi-hot feature vectors that share active slots across items are unit-norm but not orthogonal, so the orthonormal formula no longer maintains the invariant. We adopt the generalised formula $\rho = \sqrt{1 + \beta^2(\hat{c}^2 - 1)} - \beta \hat{c}$ (where $\hat{c} = \mathbf{c}_{\text{prev}} \cdot \mathbf{c}^{\text{IN}}$) from Polyn et al.^{S4} Eq. 3, which reduces exactly to $\sqrt{1 - \beta^2}$ when $\hat{c} = 0$ and preserves unit norm for arbitrary unit-norm inputs. This is applied at every drifting context in MS-TCM: the per-storyline level (rate β_{story} on storyline- s items only), the global level (rate β_{enc}^G on every step), and the retrieval-cue drifts (rate β_{rec}). See `code/ms_tcm/drift.py::norm_preserving_rho`.

Softmax gain k . The retrieval readout $a_j = M^{\text{IC}^\top} \mathbf{c}^{\text{ret}}$ produces activations whose range is bounded; the softmax gain $k > 0$ sharpens the distribution. Without k , the softmax $P(j) \propto \exp(a_j)$ over bounded activations is close to uniform and the model fails to produce recency or contiguity

signatures. k is fit alongside the other parameters; it is initialized at the Cornell and Zhang (2025) Table 1 value $k = 6.50$.

Primacy gradient. Standard CMR inherits a Polyn-et-al.-style primacy scaling $p(t) = 1 + \phi \exp(-\psi t)$ applied to the M^{IC} update magnitude at encoding step t . This is a v6-native CMR mechanism, not an MS-TCM extension. Under the `--standard-tcm` reduction the hierarchical layer is disabled and the implementation falls back to a stronger primacy gradient ($\phi = 30.0$, $\psi = 0.8$) to keep the shape assertion satisfied; under MS-TCM the per-storyline associative matrices contribute additional primacy through the storyline-level beginning-of-list synchronization at the start of each route- β visit, so a modest gradient ($\phi = 1.5$, $\psi = 0.5$) is sufficient.

No-repeats candidate set. Participants almost never repeat an item within one recall sequence (intra-list repeat rates below 1%). A softmax readout without an exclusion constraint, however, would predict the most recent item with near-certainty whenever k is moderate. We therefore mask the softmax output to zero for any serial position already recalled on the current list before re-normalising — the standard CMR convention. The likelihood routine skips observed repeats (rather than returning $-\infty$) so the handful of real repeats in the data do not poison the dataset log-likelihood.

S4. Numerical-accuracy anchors and model comparison

The numerical anchor for MS-TCM’s implementation has two parts. First, the Cornell and Zhang^{S1} hierarchical free-recall implementation in `code/ms_tcm/_likelihood_core.py` is verified to 10^{-10} tolerance against hand-derived oracle tests at `code/tests/test_hcmr_oracle.py`, so the comparator against which MS-TCM’s ΔLL is reported is itself a faithful implementation of its published inheritance. Second, when MS-TCM is configured with $\lambda = 0$, $\tau = 0$, and a single storyline ($K = 1$), it reduces exactly to that C&Z baseline at the same machine-epsilon tolerance — a regression covered by `code/tests/test_hcmr_standard_tcm.py`. The Layer 1 shape assertions of `code/tests/test_behavioral_regression.py` pin the canonical free-recall signatures (SPC, pFR, lag-CRP shapes) for both the C&Z baseline and MS-TCM. Together these anchors establish that any reported ΔLL between MS-TCM and the C&Z baseline is attributable to the new mechanisms (per-storyline contexts, the global cross-storyline context, per-storyline associative matrices, λ , the τ -mixture between recency-driven global recall and strict hierarchical recall) rather than to implementation drift.

Supplementary references

- [S1] Charlotte A. Cornell and Qiong Zhang. Hierarchical context guides human memory search. *Psychological Review*, 2025. Preprint in `notes/CornZhan25.pdf`; DOI to be assigned upon publication.
- [S2] M W Howard and M J Kahana. A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46:269–299, 2002.
- [S3] Jeremy R Manning, Emily C Whitaker, Paxton C Fitzpatrick, Madeline R Lee, Allison M Frantz, Bryan J Bollinger, Darya Romanova, Campbell E Field, and Andrew C Heusser. Feature and order manipulations in a free recall task affect memory for current and future lists, 2023. PsyArXiv preprint, <https://psyarxiv.com/erzfp>. Data and code: <https://github.com/ContextLab/FRFR-analyses>.
- [S4] S M Polyn, K A Norman, and M J Kahana. Task context and organization in free recall. *Neuropsychologia*, 47:2158–2163, 2009.